HWA CHONG INSTITUTION
JC2 Preliminary Examination
Higher 2

CANDIDATE NAME		**CT GROUP**	
CENTRE NUMBER		**INDEX NUMBER**	

GEOGRAPHY

9173/01

Paper 1

29 August 2025

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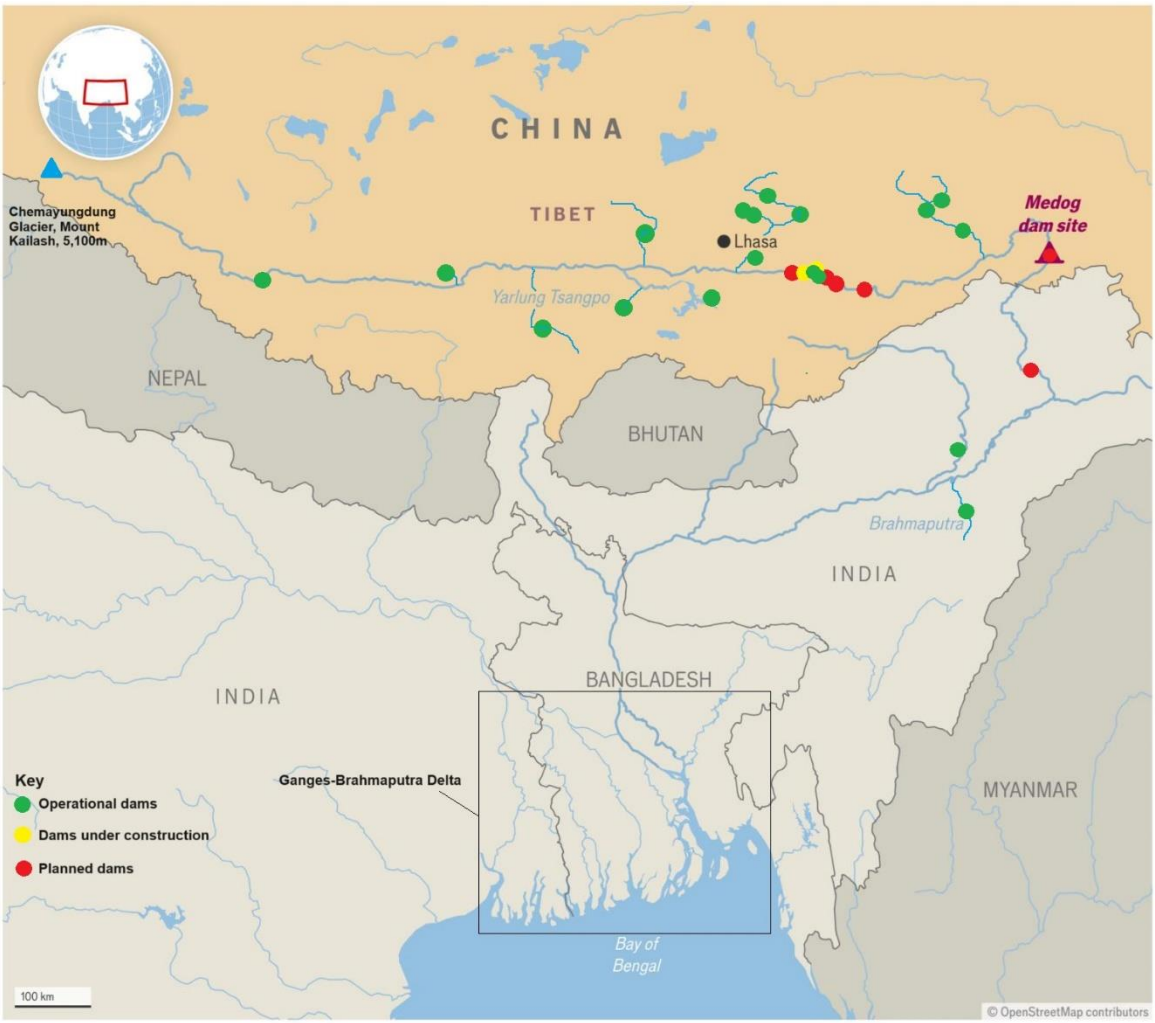
3 hours

READ THESE INSTRUCTIONS FIRST

This Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

Major dam projects along Yarlung Tsangpo (China) – Brahmaputra (India and Bangladesh) and associated tributaries



Source: The Economist.

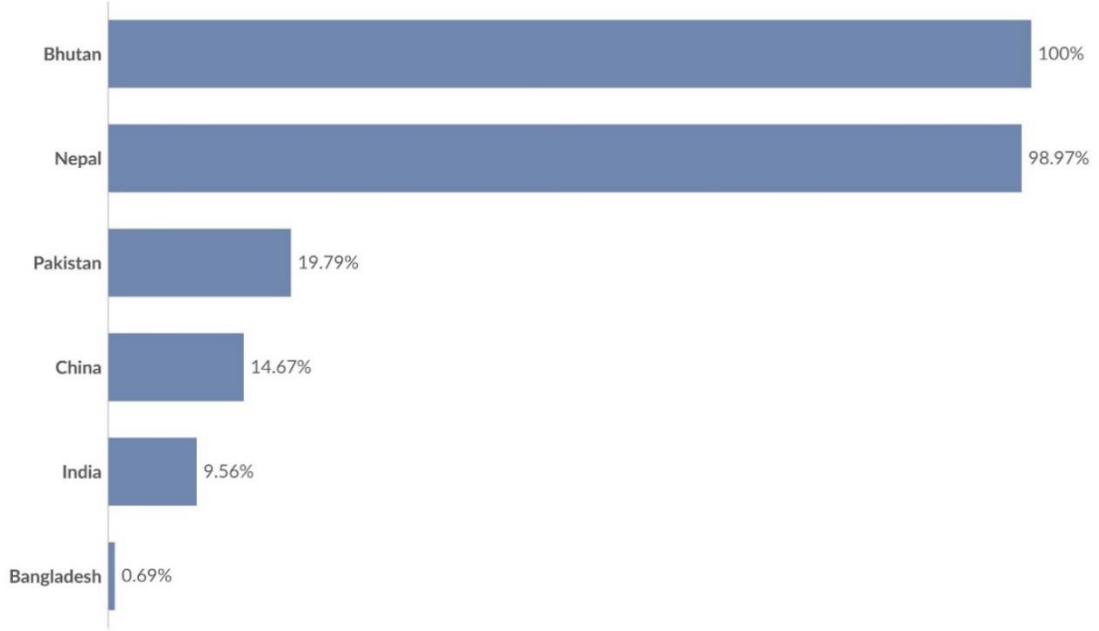
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China is planning a massive, 60GW-capacity super dam in Tibet – the Medog Dam – to produce triple the electricity generated by the world's largest power station, The Three Gorges.

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Resource 2 for Question 1

Share of electricity production from hydropower, 2022
Measured as a percentage of total electricity produced in the country



Resource 3 for Question 1

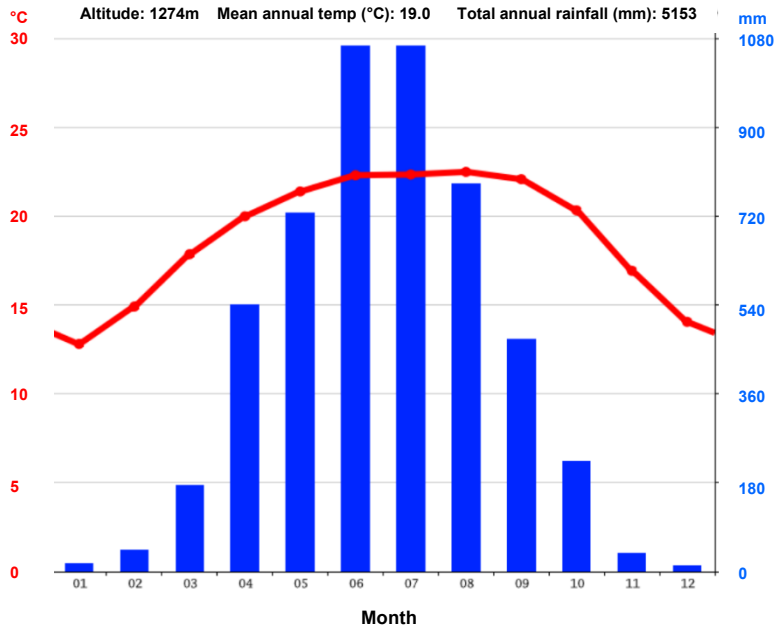
Fact File on China, India and Bangladesh

	China	**India**	**Bangladesh**
Population relying on Yarlung Tsangpo – Brahmaputra River for water	Unknown	39 million	58 million
% Population relying on Yarlung Tsangpo – Brahmaputra River for water	Unknown	3%	35%
% of discharge of Yarlung Tsanpo – Brahmaputra River which is generated from country	~18% Mostly through snow and glacial melt in upper Tibetan reaches	>73.9% Mostly through monsoon rainfall and tributaries	<8.1% Through rainfall and tributaries

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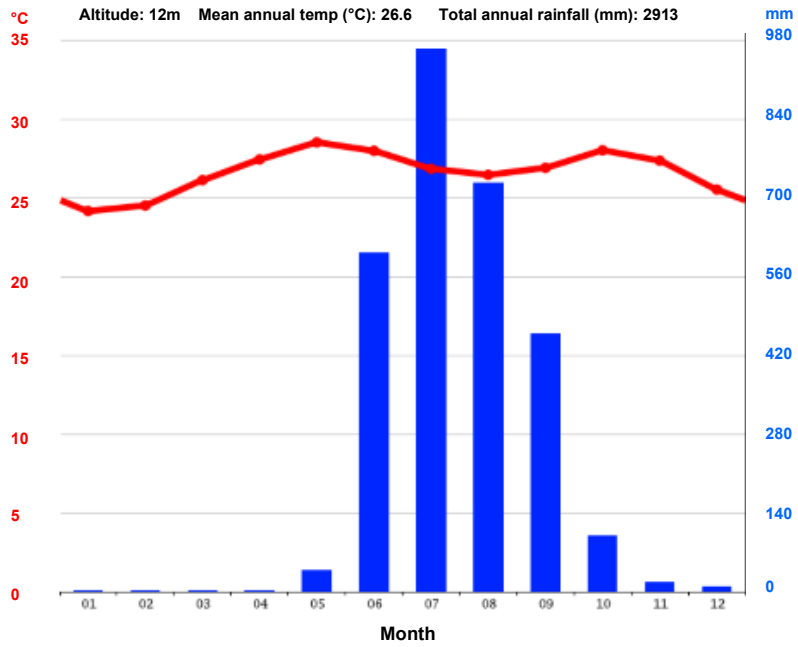
Resource 4A for Question 2

Climograph for Cherrapunji, India



Resource 4B for Question 2

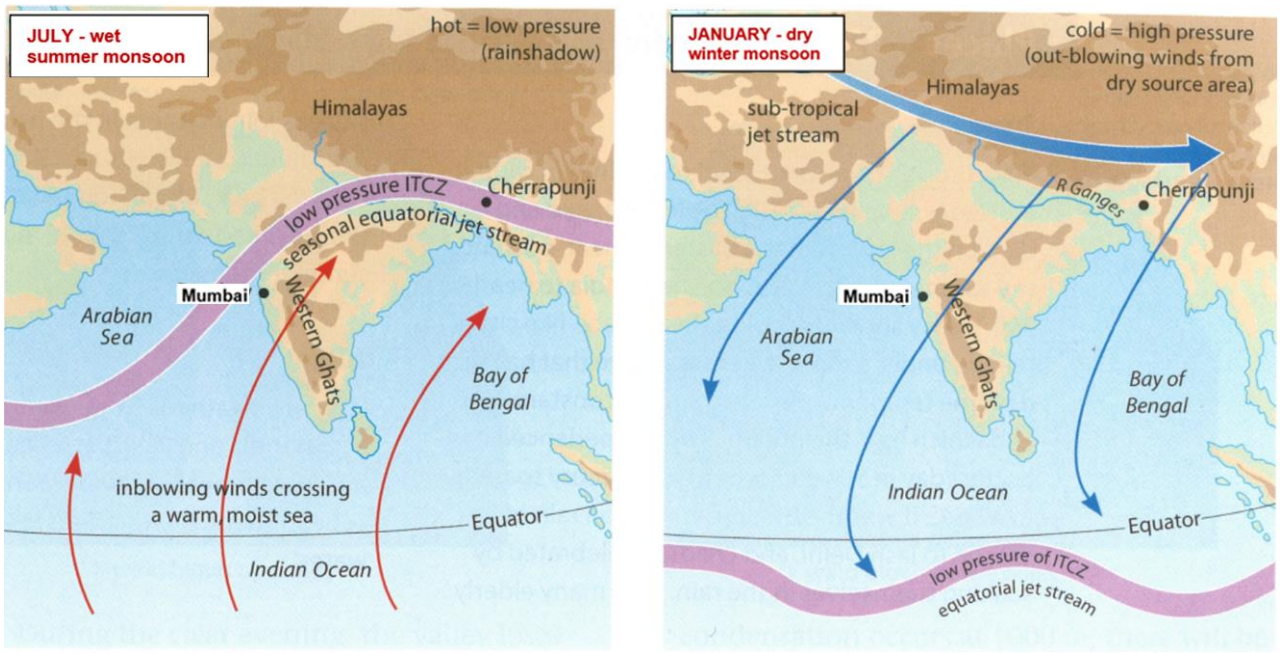
Climograph for Mumbai, India



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Resource 5 for Question 2

Monsoon circulation over India in July (left) and January (right)



Resource 6 for Question 2

Structural characteristics of a 'monsoon low' and tropical cyclone

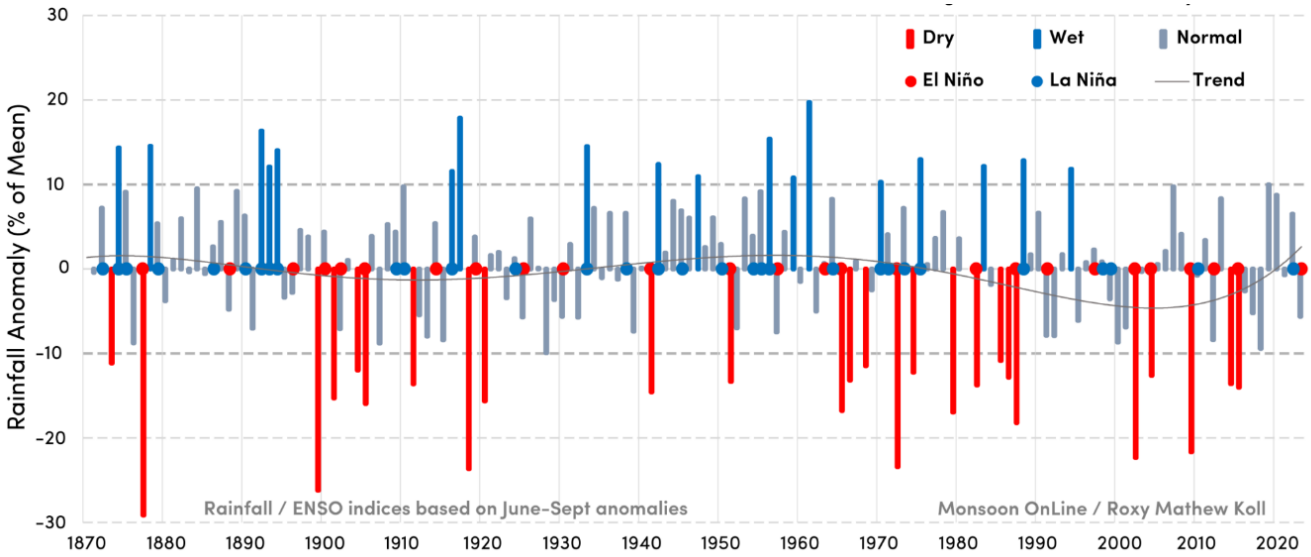
Structural characteristics	Monsoon Low	Well-developed Tropical Cyclone
Rotating circulation	May have more than one centre	Has only one centre
The region of strongest winds	Usually along the periphery	Concentrated near the centre
Wind speed near the centre	Relatively low	Strongest
Cloud clusters near the centre	Developing and dissipating without much organisation	Intense and persistent; and organized in spirals

Monsoon low/trough: the section of ITCZ from the South China Sea to the Indian Ocean

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Resource 7 for Question 2

India's summer monsoon (wet monsoon) rainfall (1871–2023) and El Niño and La Niña years



Note:
Drought years (rainfall anomaly of < -10%) are marked in red color
Wet years (rainfall anomaly of >10%) are marked in dark blue color
El Niño and La Niña conditions for the monsoon season are marked using red and blue dots

Acknowledgements:
Question 1 Resource 1 @ https://www.rfa.org/english/news/china/tibet-dam-12032020171138.html
Question 1 Resource 2 @ https://ourworldindata.org/
Question 1 Resource 3 @ https://www.preventionweb.net/news/choking-brahmaputras-flow-much-ado-about-nothing
Question 2 Resource 4 @ https://en.climate-data.org/asia/india/
Question 2 Resource 5 @ https://licchavilyceum.com/monsoon/
Question 2 Resource 6 @ https://www.hko.gov.hk/en/education/weather/monsoons/00491-monsoon-lows-and-tropical-cyclones.html
Question 2 Resource 7 @ https://www.climate.rocksea.org/research/enso-monsoon/